

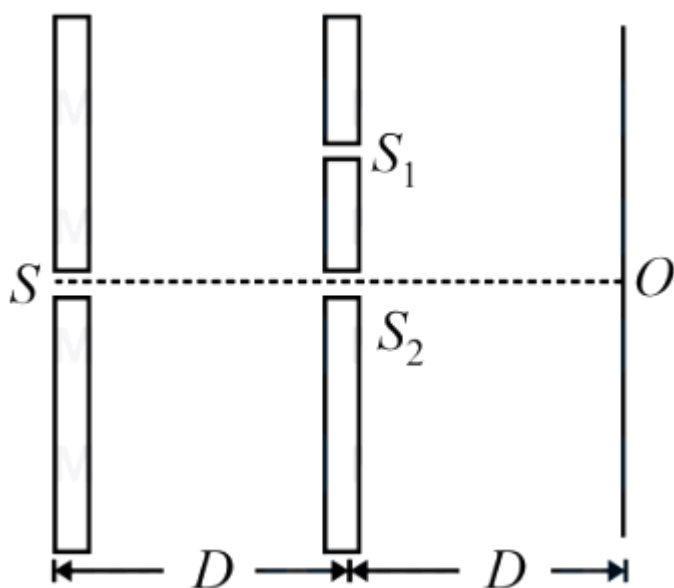
Q1

A particle of mass ' m ' is rotating in a circle of radius ' r ' with power $P = mk^2r^2t$. If the particle starts its motion from rest, then its centripetal acceleration is :

- (1) kr^2t^2
- (2) k^2r^2t
- (3) k^2rt^2
- (4) kr^2t

Q2

Two ideal slits S_1 and S_2 are at a distance d apart and illuminated by light of wavelength λ , passing through an ideal source slit S , placed on the line through S_2 , as shown. The distance between the planes of slits and the source slit is D . A screen is held at a distance D from the plane of the slits. The minimum value of d for which there is darkness at O is



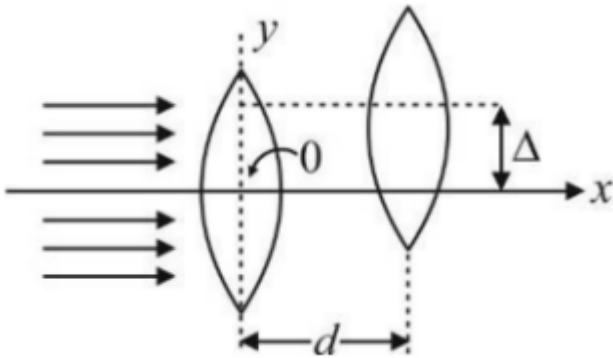
- (1) $\sqrt{\frac{3\lambda d}{2}}$
- (2) $\sqrt{\lambda D}$
- (3) $\sqrt{\frac{\lambda D}{2}}$
- (4) $\sqrt{3\lambda D}$

Questions with Answer Keys

MathonGo

Q3

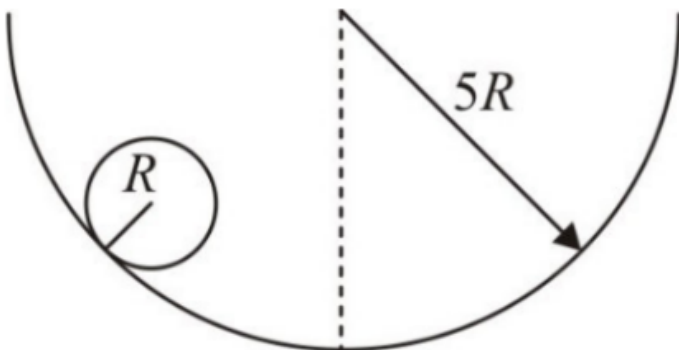
Two thin convex lenses of focal lengths f_1 and f_2 are separated by a horizontal distance d (where $d < f_1$ and $d < f_2$) and their centres are displaced by a vertical separation Δ as shown. Taking the origin of coordinates O at the centre of first lens, x and y coordinates of the focal point of this lens system, for a parallel beam of rays coming from the left, are given by: (Assume that $\Delta \ll f$)



- (1) $x = \frac{f_1 f_2}{f_1 + f_2}, y = \Delta$
- (2) $x = \frac{f_1(f_2 + d)}{f_1 + f_2 - d}, y = \frac{\Delta}{f_1 + f_2}$
- (3) $x = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}, y = \frac{\Delta(f_1 - d)}{f_1 + f_2 - d}$
- (4) $x = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}, y = 0$

Q4

A ring of mass m and radius R rolls without slipping on a fixed cylindrical surface of radius $5R$ as shown in the figure. The time period of small oscillation of ring is :



- (1) $2\pi\sqrt{\frac{2R}{g}}$
- (2) $2\pi\sqrt{\frac{8R}{g}}$

Questions with Answer Keys

MathonGo

(3) $\pi\sqrt{\frac{2R}{g}}$

(4) None of these

Q5

A large insulating thick sheet of thickness $2d$ carries a uniform charge per unit volume ρ . A particle of mass m , carrying a charge q having a sign opposite to that of the sheet, is released from the surface of the sheet. The sheet does not offer any mechanical resistance to the motion of the particle. Find the oscillation frequency ν of the particle inside the sheet.

(1) $\nu = \frac{1}{2\pi} \sqrt{\frac{q\rho}{m\epsilon_0}}$

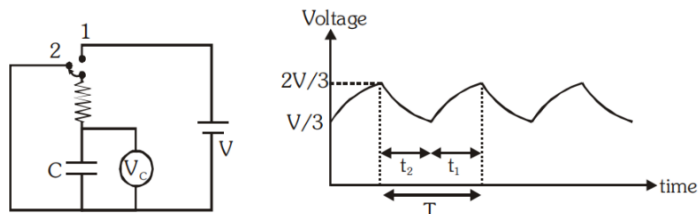
(2) $\nu = \frac{1}{2\pi} \sqrt{\frac{2q\rho}{m\epsilon_0}}$

(3) $\nu = \frac{1}{4\pi} \sqrt{\frac{q\rho}{m\epsilon_0}}$

(4) None of these

Q6

The switch in circuit shifts from 1 to 2 when $V_C > 2V/3$ and goes back to 1 from 2 when $V_C < V/3$. The voltmeter reads voltage as plotted. What is the period T of the wave form in terms of R and C ?



(1) $RC \ln 3$

(2) $2RC \ln 2$

(3) $\frac{RC}{2} \ln 3$

(4) $\frac{RC}{3} \ln 3$

Q7

Questions with Answer Keys

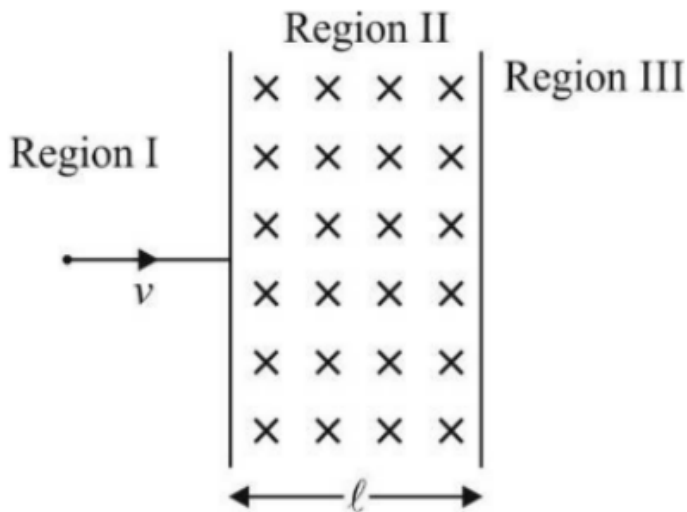
MathonGo

A spherical planet of radius R has spherically symmetrical distribution of mass density, varying as square of the distance from the centre, from zero at centre to maximum value ρ_0 at its surface.

- (1) The value of escape velocity of a mass m at the surface of planet is $\sqrt{\frac{4\pi G_0 R^2}{5}}$.
- (2) The value of acceleration due to gravity ' g ' varies inside the planet as cube of the distance from centre.
- (3) The value of escape velocity from surface is same as the escape velocity from another planet of same total mass & radius but having uniform mass density.
- (4) The energy required to impart escape velocity to particles of masses ' m ' & ' $2m$ ', at the surface of planet, will be in ratio 1 : 2.

Q8

A particle of mass m and charge q moving with a velocity V enters region II normal to the boundary as shown in figure. Region II has a uniform magnetic field B perpendicular to the plane of paper and length of region II is ℓ . Choose the correct choice.



- (1) Particle enters region III only if its velocity $v > \frac{B\ell q}{m}$
- (2) Particle enters region III only if its velocity $v < \frac{B\ell q}{m}$
- (3) Path length of particle in region II is maximum when $v = \frac{B\ell q}{m}$
- (4) Time spent in region II is same for any velocity as long as particle returns to region I

Q9

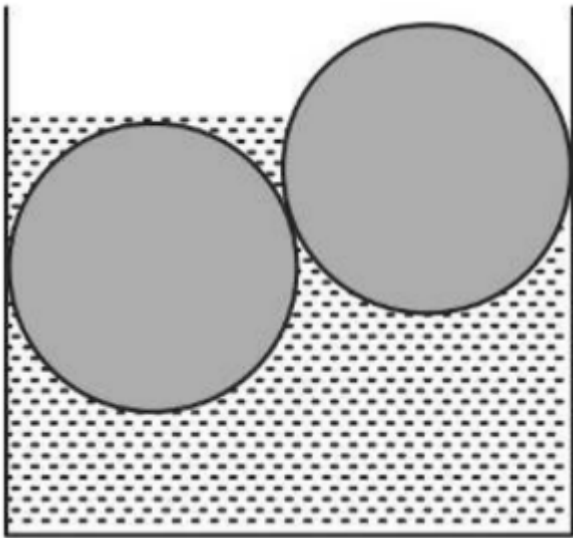
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<https://www.mathongo.com>

Questions with Answer Keys

MathonGo

There are two cylinder-shaped wooden billets, each having a mass of 45 kg, in a vertical wall sewage, which contains water in it. The two billets have the same size and the same material; they touch each other and the walls of the sewage. One of them is totally under the water, while only half of the other one is immersed into the water. Friction is negligible everywhere. (Density of water is 1000 kg/m^3 and $g = 10 \text{ m/s}^2$). Choose the correct option(s).

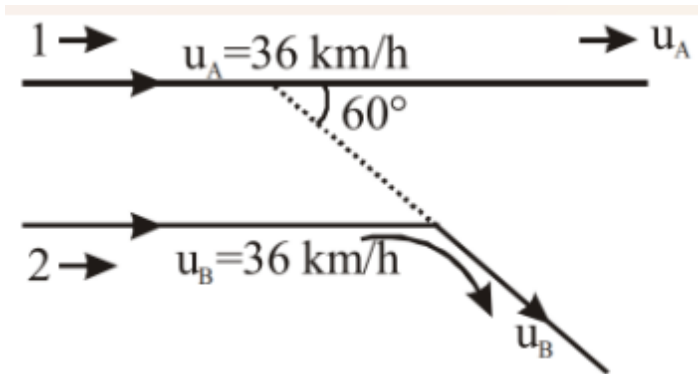


- (1) Density of the wood is 750 kg/m^3
- (2) Density of the wood is 800 kg/m^3
- (3) The forces exerted by the billets on the vertical walls is 260 N approximately
- (4) The forces exerted by the billets on the vertical walls is 320 N approximately

Q10

Two trains move in the same direction on two close tracks. The train A sounds a horn of single frequency of 1 kHz. At a certain instant of time, when both trains move at the same speed of 36 km/hr, the two straight tracks deviate at an angle of 60° with each other. Velocity of sound in air is 300 m/s. Which of the following

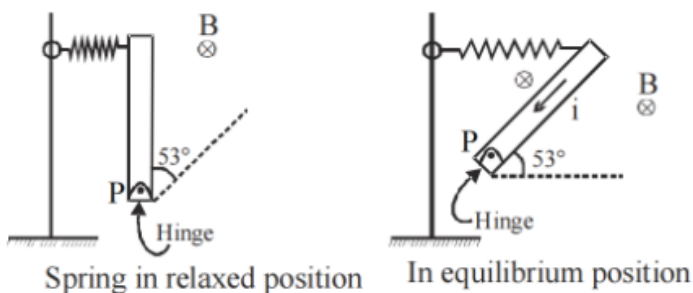
statement(s) is/are true ?



- (1) The apparent frequency heard by a passenger on the second train B just after he passes the bend will be 1.05kHz.
- (2) The apparent frequency heard by a passenger on the first train A, just after the second train B passes the bend will be 1kHz.
- (3) The apparent frequency heard by a passenger on the second train B just after he passes the bend will be 920 Hz.
- (4) If the first train A, that produces the sound, were to pass the bend while the other train B goes on the straight track then the apparent frequency heard by a passenger on the latter train B will be less than 1kHz.

Q11

A thin uniform rod with negligible mass and length ℓ is attached to the floor by a frictionless hinge at point P. A horizontal spring with force constant k connects the other end to wall. The rod is in a uniform magnetic field B directed into the plane of paper. What is extension in spring in equilibrium when a current i is passed through the rod in direction shown. Assuming spring to be in natural length initially.



- (1) $\frac{5i\ell B}{8k}$

Questions with Answer Keys

MathonGo

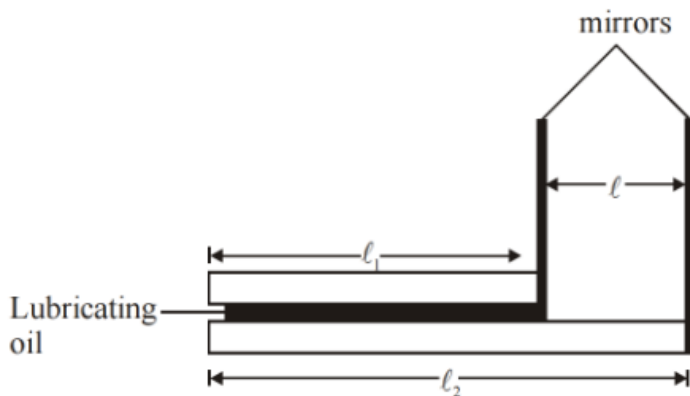
(2) $\frac{3i\ell B}{8k}$

(3) $\frac{5i\ell B}{4k}$

(4) $\frac{5i\ell B}{6k}$

Q12

An optical engineering firm needs to ensure that the separation between two mirrors is unaffected by temperature changes. The mirrors are attached to the ends of two bars of different materials that are welded together at one end as shown in figure. The surfaces of the bars in contact are lubricated. The distance ℓ does not change with temperature change. ℓ_1 and ℓ_2 are the length of the bars α_1 and α_2 are the respective thermal coefficients of temperature. Which of the following options is/are correct :



(1) $\ell_1 = \frac{\ell \alpha_2}{\alpha_1 - \alpha_2}$

(2) $\ell_2 = \frac{\ell \alpha_1}{\alpha_2 - \alpha_1}$

(3) $\alpha_1 \ell_1 = \alpha_2 \ell_2$

(4) $\alpha_1 \ell_2 = \alpha_2 \ell_1$

Q13

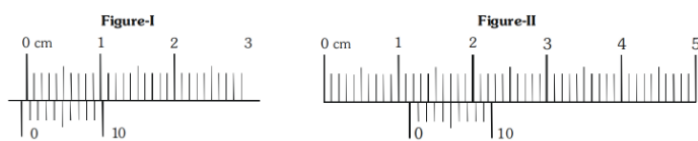
In ordinary Vernier calipers, 10^{th} division of the Vernier scale coincides with 9^{th} division of the main scale.

In a specially designed Vernier calipers the Vernier scale is so constructed that 10^{th} division on it coincides with 11^{th} division on the main scale. Each division on the main scale equals to 1 mm. The calipers have a zero error as shown in the figure-I. When the Vernier caliper is used to measure a length, the concerned portion

Questions with Answer Keys

MathonGo

of its scale is shown in figure-II.



- (1) Zero error in the calipers has magnitude 0.7 mm.
- (2) The length being measured is 1.08 cm.
- (3) The length being measured is 1.22 cm.
- (4) Though the given Vernier scale does not follow the principle of Vernier scale, yet can be used satisfactorily.

Q14

In an X-ray tube, electrons emitted from a filament (cathode) carrying current I hit a target (anode) at a distance d from the cathode. The target is kept at a potential V higher than the cathode resulting in emission of continuous and characteristic X-rays. If the filament current I is decreased to $\frac{I}{2}$, the potential difference V is increased to $2V$, and the separation distance d is reduced to $\frac{d}{2}$, then

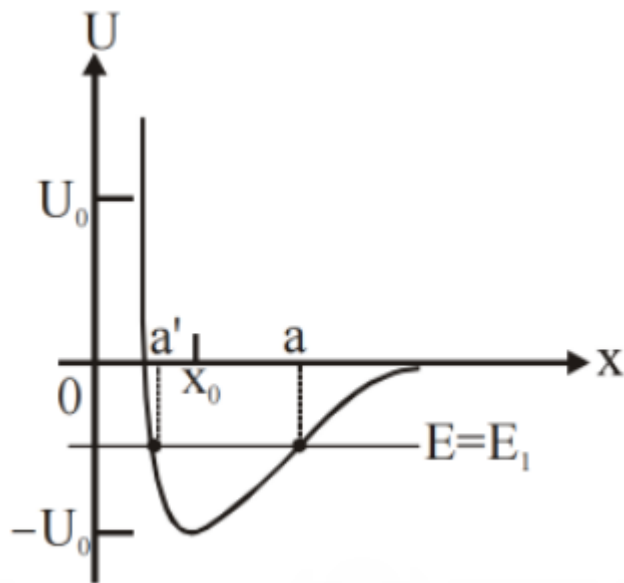
- (1) the cut-off wavelength will reduce to half, and the wavelengths of the characteristic X-rays will remain the same
- (2) the cut-off wavelength as well as the wavelengths of the characteristic X-rays will remain the same
- (3) the cut-off wavelength will reduce to half, and the intensities of all the X-rays will decrease
- (4) the cut-off wavelength will become two times larger, and the intensity of all the X-rays will decrease

Q15

In a diatomic molecule the atoms can move relative to one another within certain limits. According to a simple theory of interatomic forces, the potential energy for the motion of each atom is

$$U(x) = U_0 \left(e^{-2(x-x_0)/b} - 2e^{-(x-x_0)/b} \right)$$

Here $e = 2.718 \dots$; U_0 , x_0 and b are constants; and x is the distance from the one atom to the midpoint of the molecule.



According to eq. $v^2 = \frac{2}{m}[E - U(x)]$, v^2 is directly proportional to $E - U(x)$; thus, v^2 is large wherever the difference between E and $U(x)$ is large. We can therefore gain some insights into the qualitative features of the motion by drawing a graph of potential energy on which it is possible to display the difference between E and $U(x)$. For example, Figure shows the curve of potential energy for an atom in a diatomic molecule. On this graph, we indicate the value of the energy of the particle by a horizontal line. We will call this horizontal line the energy level of the particle. At any point x , we can then see the difference between E and $U(x)$ at a glance.

Mark the incorrect statement.

- (1) For the given graph equilibrium of molecule exists where slope of graph is zero.
- (2) For $x > x_0$ interatomic force is attractive in nature.
- (3) For $x < x_0$ interatomic force is repulsive in nature.
- (4) As particle moves away from equilibrium position, magnitude of intermolecular force decreases.

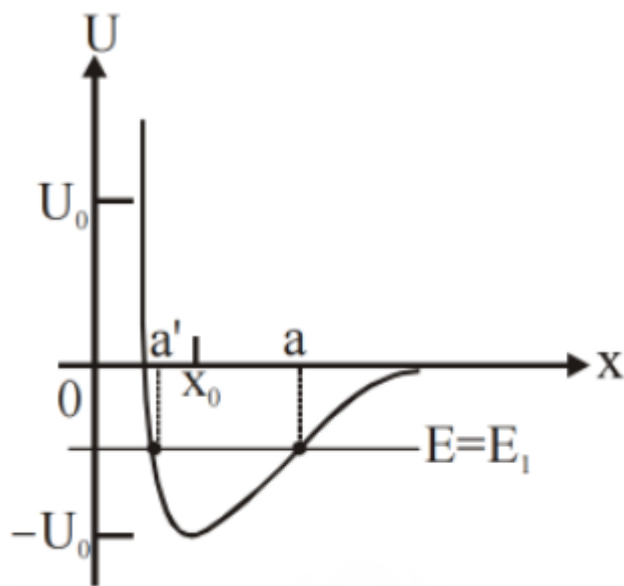
Q16

In a diatomic molecule the atoms can move relative to one another within certain limits. According to a simple theory of interatomic forces, the potential energy for the motion of each atom is

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On this graph, we indicate the value of the energy of the particle by a horizontal line. We will call this horizontal line the energy level of the particle. At any point x , we can then see the difference between E and $U(x)$ at a glance.

In the vicinity of equilibrium position $x = x_0$, for a small displacement, force is proportional to the displacement. What is value of proportionality constant k .

- (1) $k = \frac{3U_0}{b^2}$
- (2) $k = \frac{2U_0}{b^2}$
- (3) $k = \frac{6U_0}{b^2}$
- (4) $k = \frac{b^2}{3U_0}$

Q17

A particle of charge q and mass m is clamped rigidly at the circumference of a ring with mass m and radius R . Initially ring is in vertical plane resting on a sufficiently rough horizontal surface with charge q at the same

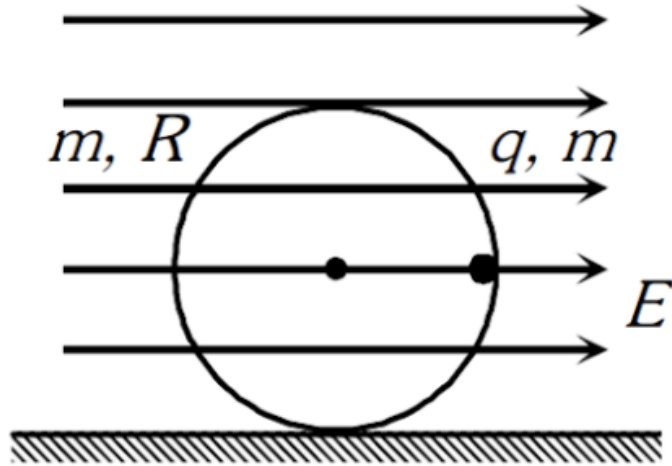
Questions with Answer Keys

MathonGo

horizontal level as that of the centre of the ring. There exists uniform horizontal electric fields as shown.

At $t = 0$ the system is let free. (Given that $qE = mg, \pi = \frac{22}{7}$)

Sufficiently rough to avoid slipping



Sufficiently rough to avoid slipping

The magnitude of the friction force at $t = 0$ is

- (1) mg
- (2) $\sqrt{2}mg$
- (3) 0
- (4) $\frac{mg}{2}$

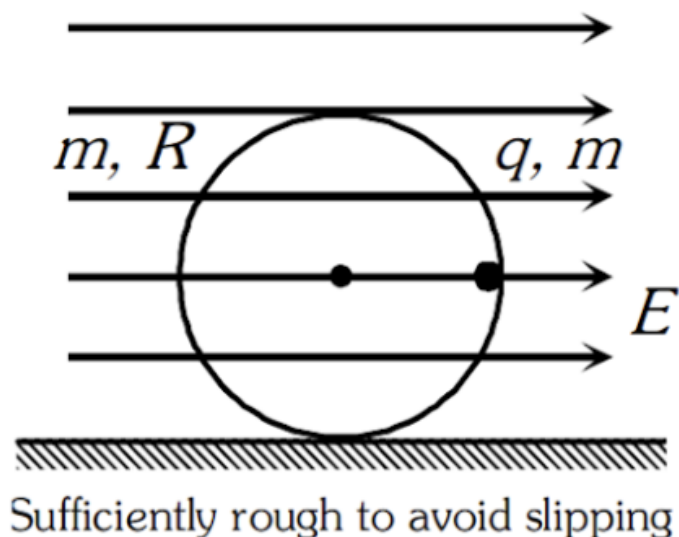
Q18

A particle of charge q and mass m is clamped rigidly at the circumference of a ring with mass m and radius R .

Initially ring is in vertical plane resting on a sufficiently rough horizontal surface with charge q at the same horizontal level as that of the centre of the ring. There exists uniform horizontal electric fields as shown.

At $t = 0$ the system is let free. (Given that $qE = mg, \pi = \frac{22}{7}$)

Sufficiently rough to avoid slipping



Work done by the electric field when the ring has rotated through 90° is

- (1) $\frac{2}{7}mgR$
- (2) $\frac{4}{7}mgR$
- (3) $-mgR$
- (4) $mg\sqrt{2}R$

Q19

H_2S is bubbled into 0.2M NaCN solution which is 0.02M in each $[Cd(CN)_4]^{2-}$ and $[Ag(CN)_2]^-$. The H_2S produces $1 \times 10^{-9}M$ sulphide in the solution.

(Given, $K_{sp}Ag_2S = 1 \times 10^{-50}M^3$; $K_{sp}CdS = 7.1 \times 10^{-28}M^2$

$K_{inst}[Ag(CN)_2]^{-1} = 1 \times 10^{-20}M^2$ $K_{ins}[Cd(CN)_4]^{-2} = 7.8 \times 10^{-18}M^4$)

Identify the correct statement

- (1) Ag_2S precipitates first from the solution
- (2) CdS precipitates first from the solution
- (3) None of them precipitate under the given conditions
- (4) Ag_2S precipitates at a sulphide concentration of $1 \times 10^{-15}M$

Q20

Questions with Answer Keys

MathonGo

Arrange the following in order of decreasing spin only magnetic moment values (in B.M.) :

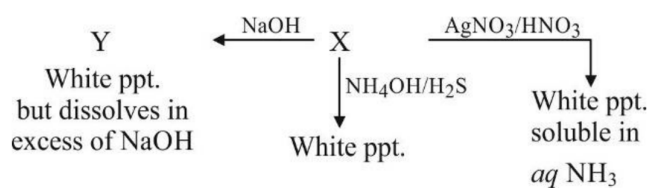
- I. $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 II. $[\text{Fe}(\text{CN})_6]^{3-}$
 III. $[\text{Fe}(\text{CN})_6]^{4-}$
 IV. $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

The correct choice is :

- (1) IV, I, II, III
 (2) I, II, III, IV
 (3) III, II, I, IV
 (4) II, III, I, IV

Q21

'X' is a colourless salt giving following reactions.

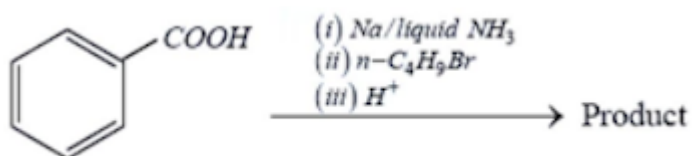


X be can.

- (1) AlCl_3
 (2) ZnCl_2
 (3) $\text{Zn}(\text{CH}_3\text{COO})_2$
 (4) ZnBr_2

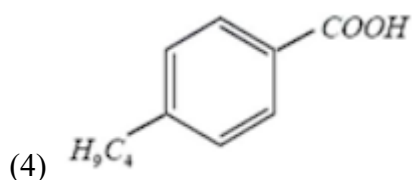
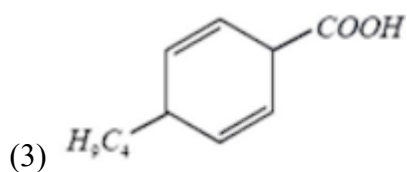
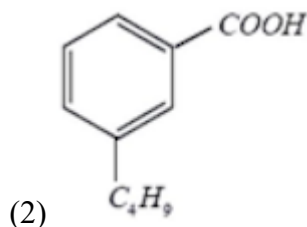
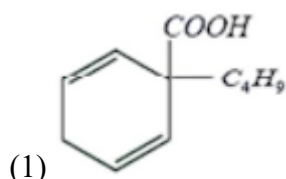
Q22

The major product formed in the following reaction is



Questions with Answer Keys

MathonGo



Q23

In paper electrophoresis, amino acids and peptides can be separated by their differential migration in an electric field. To the center of a strip of paper, wet with buffer at pH 6, is applied a mixture of the following three peptides in a single small spot : Gly-Lys, Gly-Asp, and Gly-Ala. A positively charged electrode (anode) is attached to the left side of the paper, and a negatively charged electrode (cathode) to the right side. A voltage is applied across the ends of the paper for a time, after which the peptides have separated into three spots : one near the cathode, one near the anode and one in the center, at the location of the original spot. Which statement is incorrect?

- (1) Gly-Lys migrates to the cathode
- (2) Gly-Asp migrates to the anode
- (3) Gly-Ala does not migrate
- (4) Gly-Lys is a acidic dipeptide

Q24

Questions with Answer Keys

MathonGo

For a certain orbital

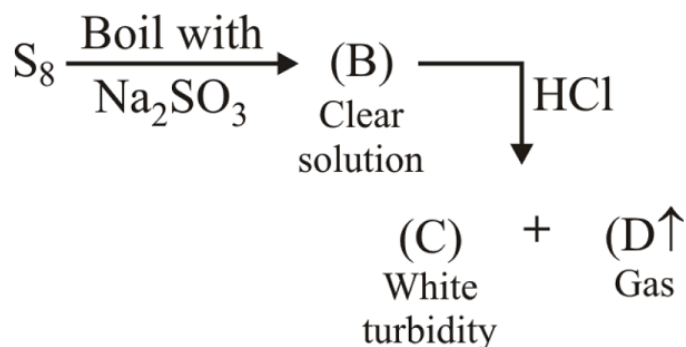
$$\Psi(r) = \frac{1}{16\sqrt{4}a_0^{3/2}} (1 - \sigma) (\sigma^2 - 8\sigma + 12) e^{-\sigma/2}, \sigma = \frac{2r}{na_0} \{a_0 = \text{constant}\}$$

The distance of nearest radial node from nucleus is xa_0 while farthest radial node from nucleus is ya_0 .

Find $\left(\frac{y}{x}\right)$.

- (1) 4
- (2) 6
- (3) 4.5
- (4) 3.5

Q25



Which of the following is/are correct statements regarding gas (D)?

- (1) Hybridisation of gas is sp^2
- (2) It can act as oxidising agent
- (3) It gives starch-iodate test
- (4) It can decolourise KMnO_4

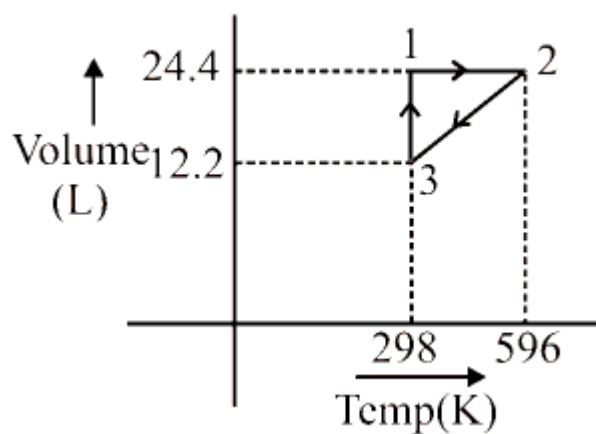
Q26

Statement-1 : Net work done by system or on the system lies between 100cal to 200cal.

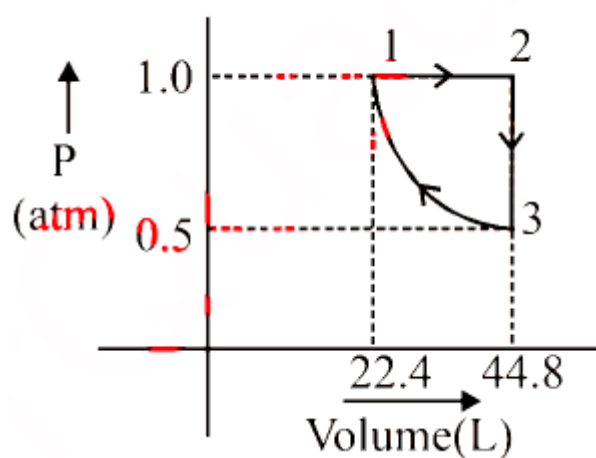
Statement-2 : Net heat released or absorbed by the system lies between 100 cal to 200cal.

Which of the following graph(s) for one mol of an ideal mono atomic gas is/are satisfying both the statements

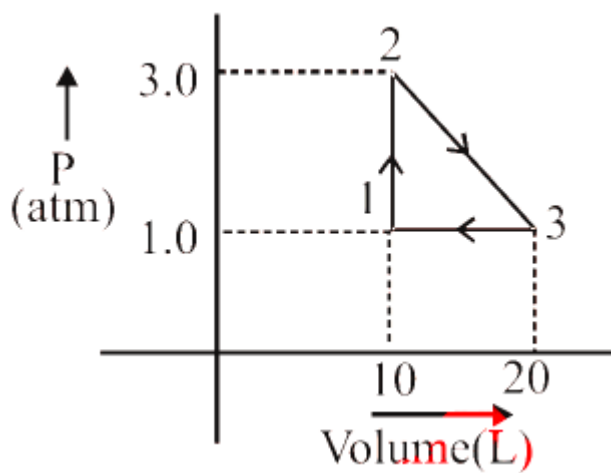
$$\left(\log 2 = 0.30, R = \frac{2\text{cal}}{\text{molK}} \right) ?$$



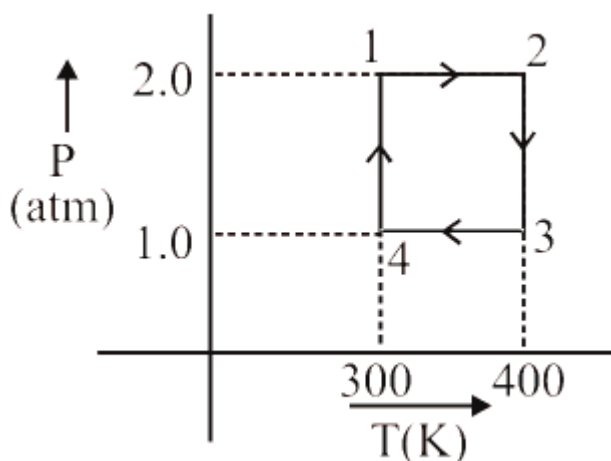
(1)



(2)



(3)



(4)

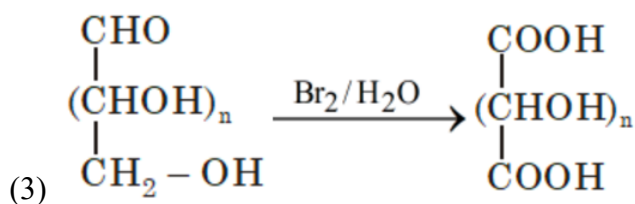
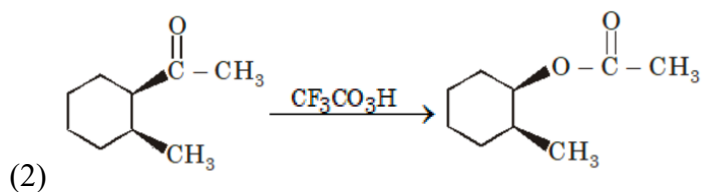
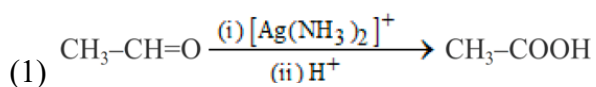
Q27

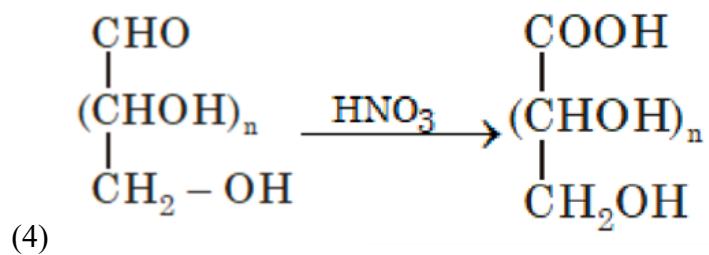
Select the correct options :

- (1) Gas is more compressible, if repulsive forces dominate over attractive forces between molecules
- (2) At extremely low pressure and high temperature, gases behave ideally
- (3) At Boyle's temperature, gases behave ideally in low pressure region
- (4) If a gas is kept at $T > T_C$, it can be liquified by lowering pressure alone

Q28

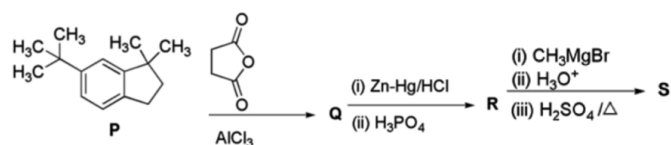
In which of the following reaction(s) the product is/are correctly matched?



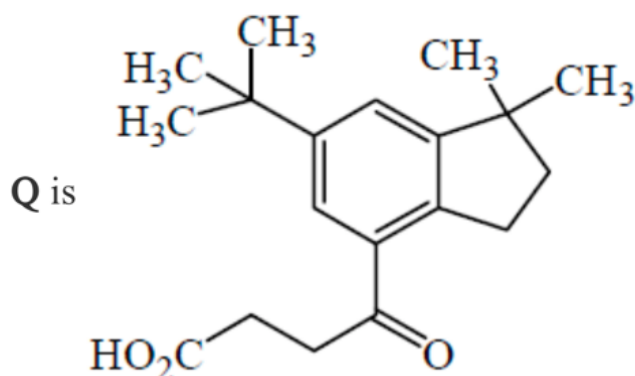
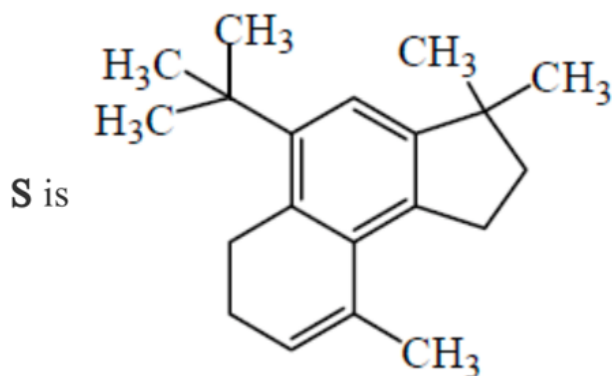


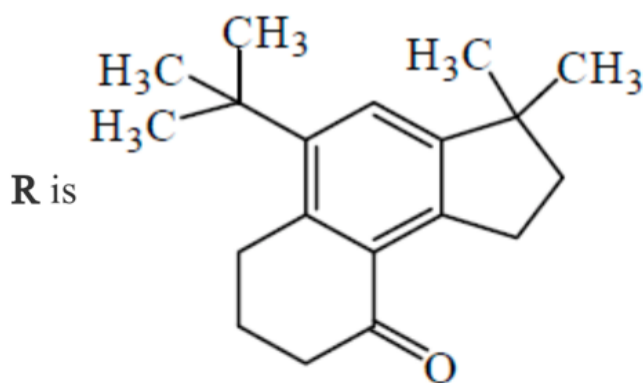
Q29

In the reaction scheme shown below **Q**, **R** and **S** are the major products.

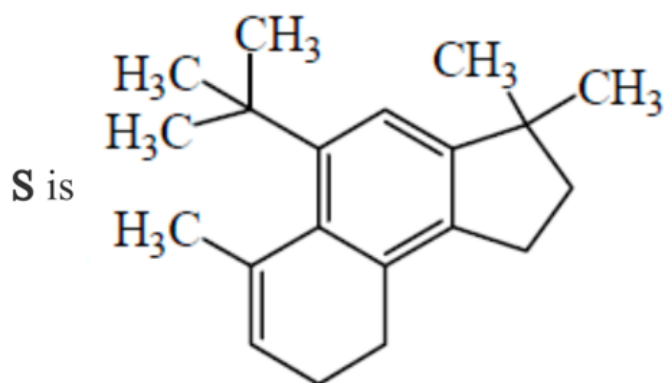


The correct structure of





(3)



(4)

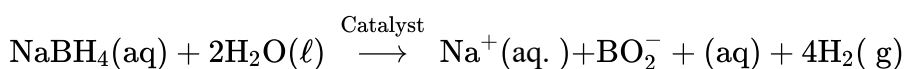
Q30

A colorless aqueous solution contains nitrates of two metals, **X** and **Y**. When it was added to an aqueous solution of NaCl, a white precipitate was formed. This precipitate was found to be partly soluble in hot water to give a residue **P** and a solution **Q**. The residue **P** was soluble in aq. NH_3 and also in excess sodium thiosulfate. The hot solution **Q** gave a yellow precipitate with KI. The metals **X** and **Y**, respectively, can be

- (1) Ag and Pb
- (2) Ag and Cd
- (3) Cd and Pb
- (4) Cd and Zn

Q31

The following reaction is a slow reaction and hence Ruthenium (molar mass = 101gm) is used as catalyst



Questions with Answer Keys

MathonGo

Kinetic studies show that reaction is 1st order with respect to catalyst but zero-order with respect to substrate.

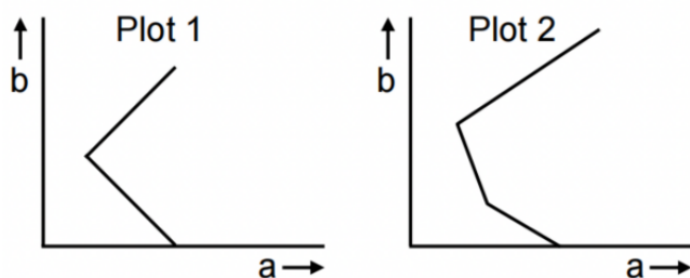
The rate of hydrogen production per mole of Ruthenium is $92 \text{ mol H}_2 (\text{mol Ru})^{-1} \text{ min}^{-1}$ at 27°C . Identify

CORRECT option(s) $\{R = 0.08 \text{ atm} - \ell/\text{mol} - \text{K}\}$

- (1) Approximately 0.045 mg Ru , must be added to 0.1 dm^3 , $1 \text{ mol dm}^{-3} \text{ NaBH}_4$ to supply H_2 gas at rate of $0.1 \text{ dm}^3 \text{ min}^{-1}$ at 27°C and 1 atm
- (2) Approximately 4.574 mg of Ru , must be added to 0.1 dm^3 , $1 \text{ mol dm}^{-3} \text{ NaBH}_4$ to supply H_2 gas at rate of $0.1 \text{ dm}^3 \text{ min}^{-1}$ at 27°C and 1 atm
- (3) If 0.1 dm^3 , $1 \text{ mol dm}^{-3} \text{ NaBH}_4$ is used to supply H_2 at at rate of $0.1 \text{ dm}^3 \text{ min}^{-1}$ at 27°C , 1 atm , then upto 96 min hydrogen can be supplied in this way
- (4) If 0.1 dm^3 , $1 \text{ mol dm}^{-3} \text{ NaBH}_4$ is used to supply H_2 at at rate of $0.1 \text{ dm}^3 \text{ min}^{-1}$ at 27°C , 1 atm , then upto 48 min hydrogen can be supplied in this way

Q32

Given below are 2 conductometric titration plots. In each plot conductance of a solution in flask is plotted on x-axis (shown as a) and the volume of the titrant added from burette is plotted on y-axis (shown as b). Which of the following option(s) is/are correct ?



- (1) Plot 1 corresponds to titration of NaOH (in flask) versus HCl (in burette)
- (2) Plot 2 corresponds to titration of NaOH (in flask) versus HCl (in burette)
- (3) Plot 1 corresponds to titration of NaOH (in flask) versus mixed acid $\text{HCl} + \text{CH}_3\text{COOH}$ (in burette)
- (4) Plot 2 corresponds to titration of NaOH (in flask) versus mixed acid $\text{HCl} + \text{CH}_3\text{COOH}$ (in burette)

Q33

MathonGo

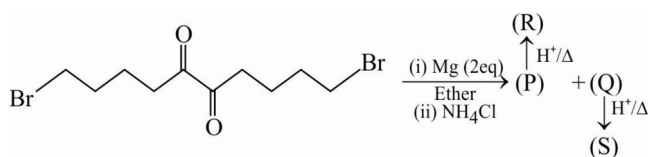
<https://www.mathongo.com>

Questions with Answer Keys

MathonGo

Paragraph:

In the following reaction sequence, if (P) can show stereoisomerism.



Question:

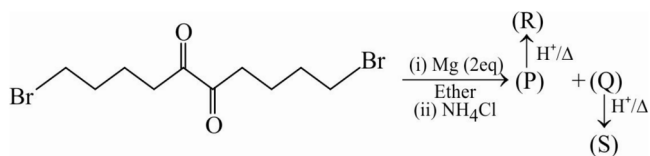
Relation between (R) and (S) is :

- (1) Homomers
- (2) Enantiomers
- (3) Diastereomers
- (4) Structural Isomers

Q34

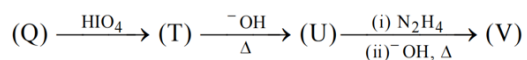
Paragraph:

In the following reaction sequence, if (P) can show stereoisomerism.



Question:

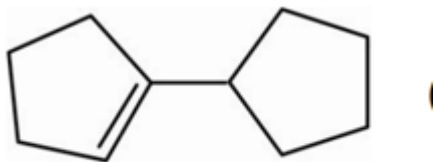
The product (V) is :



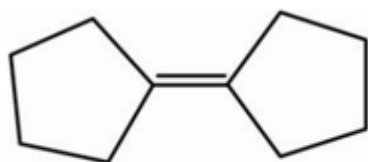
(1)

Questions with Answer Keys

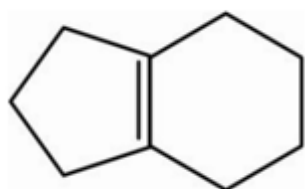
MathonGo



(2)



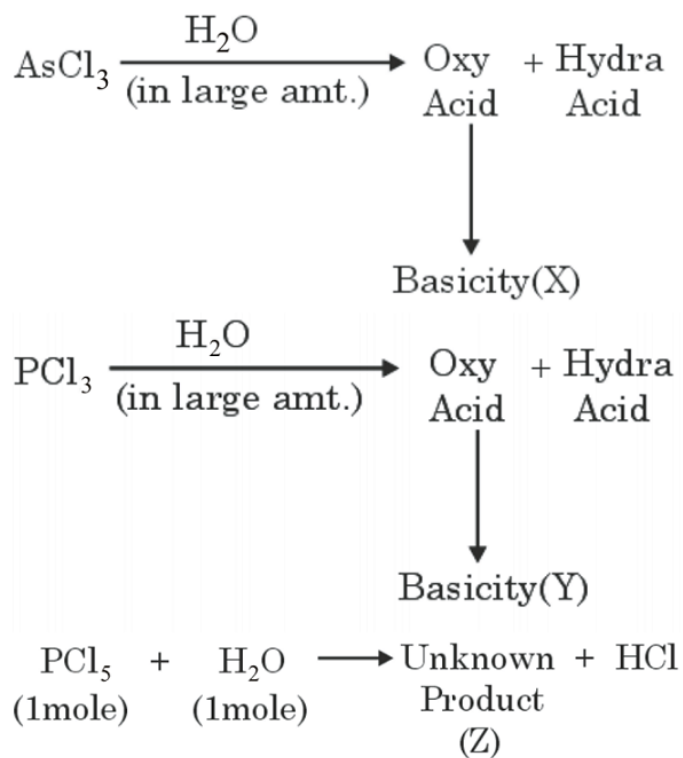
(3)



(4)

Q35

Paragraph:

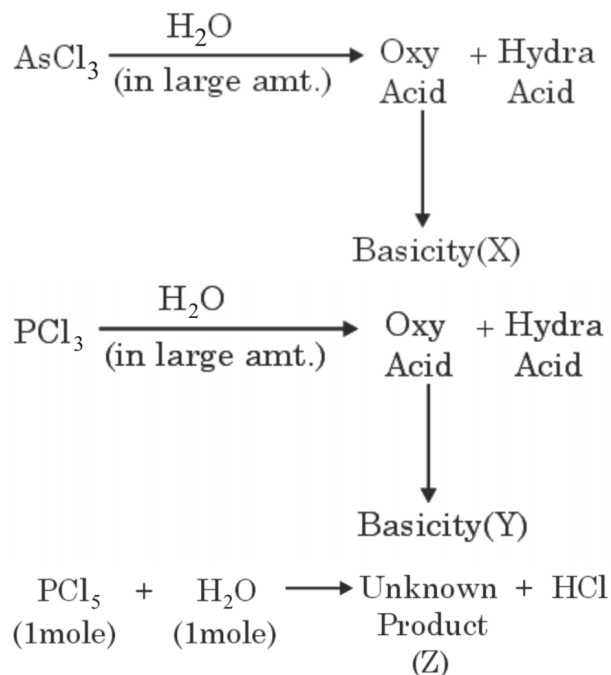


Question:

The value of $\left(\frac{X}{Y}\right)$ is.

- (1) 1.5
- (2) 2
- (3) 0.5
- (4) 0.25

Q36

Paragraph:**Question:**

The ratio of number of π bond(s) to number of σ bond(s) in ' Z ' is

- (1) 1.5
- (2) 2
- (3) 0.5
- (4) 0.25

Q37

In the XY -plane, the length of the shortest path from $(0, 0)$ to $(12, 16)$ that does not go inside the circle

$$(x - 6)^2 + (y - 8)^2 = 25, \text{ is}$$

- (1) $10\sqrt{3}$
- (2) $10\sqrt{5}$
- (3) $10\sqrt{3} + \frac{5\pi}{6}$
- (4) $10\sqrt{3} + \frac{5\pi}{3}$

Questions with Answer Keys

MathonGo

Q38

$\lim_{x \rightarrow 0} \frac{\sin x^4 - x^4 \cos x^4 + x^{20}}{x^4(e^{2x^4} - 1 - 2x^4)}$ is equal to

- (1) 0
- (2) $-\frac{1}{6}$
- (3) $\frac{1}{6}$
- (4) does not exist

Q39

The number of positive integer solutions of $a + b + c = 60$, where a is a factor of b and c , is

- (1) 184
- (2) 200
- (3) 144
- (4) 270

Q40

Let us define a region R in xy -plane as a set of points (x, y) satisfying $[x^2] = [y]$ (where $[x]$ denotes greatest integer $\leq x$), then the region R defines

- (1) a parabola whose axis is horizontal
- (2) a parabola whose axis is vertical
- (3) integer point of the parabola $y = x^2$
- (4) None of the above

Q41

If $f(x)$ be a function such that $f(x+1) = \frac{f(x)-1}{f(x)+1}$, $\forall x \in \mathbb{N}$ and $f(1) = 2$ then $f(999)$ is -

- (1) -3
- (2) 2

Questions with Answer Keys

MathonGo

(3) $\frac{1}{3}$

(4) $-\frac{1}{2}$

Q42

Let $f(x)$ ($x \geq 1$) be a differentiable function satisfying

$$f(x) = (\log_e x)^2 - \int_1^e \frac{f(t)}{t} dt$$

Area bounded by tangent line of $y = f(x)$ at the point $[e, f(e)]$, the curve $y = f(x)$ and the line $x = 1$, is :

(1) $e + \frac{1}{e}$

(2) $e + \frac{1}{e} - 1$

(3) $e + \frac{1}{e} - 2$

(4) $e + \frac{1}{e} - 3$

Q43

If $\left| \frac{z-\alpha}{z-\beta} \right| = k, k > 0$ where, $z = x + iy$ and $\alpha = \alpha_1 + i\alpha_2, \beta = \beta_1 + i\beta_2$ are fixed complex numbers. Then which of the following are true

(1) if $k \neq 1$ then locus is a circle whose centre is $\left(\frac{k^2\beta - \alpha}{k^2 - 1} \right)$

(2) if $k \neq 1$ then locus is a circle whose radius is $\left| \frac{k(\alpha - \beta)}{1 - k^2} \right|$

(3) if $k = 1$ then locus is perpendicular bisector of line joining $\alpha = \alpha_1 + i\alpha_2$ and $\beta = \beta_1 + i\beta_2$

(4) if $k \neq 1$ then locus is a circle whose centre is $\left(\frac{k^2\alpha - \beta}{k^2 - 1} \right)$

Q44

Suppose that M is a natural number with the property that if x is chosen randomly from the set

$\{1, 2, \dots, 1000\}$ the probability that x is divisor of M is $\frac{1}{2}$. If $M \leq 1000$ then maximum possible value of

M is divisible by -

(1) 2

(2) 61

(3) 3

(4) 5

Q45

If a_1, a_2, a_3, a_4, a_5 are distinct positive terms in AP having common difference d , then -

(1) $5a_3^2 > 4d^2$

(2) sum of all terms $= 5a_3$

(3) $a_1 + 5a_5, 3a_3, 2a_2 + 4a_4$ are in A.P.

(4) $a_1a_5 < a_2a_4$

Q46

Assume that $A_i (i = 1, 2, \dots, n)$ are the vertices of a regular polygon inscribed in a circle of radius unity. The value of $\frac{|A_1A_2|^2 + |A_1A_3|^2 + \dots + |A_1A_n|^2}{n}$ is:

(1) > 1.5

(2) 1

(3) 2

(4) < 1.5

Q47

Let $f(x)$ be a monic polynomial of degree 4 satisfying the following conditions:

(i) $f'(0) = 0$

(ii) $f'(2) = 16$

(iii) for some positive real k , $f'(x) < 0$ in the intervals $(-\infty, 0)$ and $(0, k)$. Then which of the followings is/are correct?

(1) Equation $f'(x) = 0$ has one real root in the interval $(0, 2)$.

(2) Function $f(x)$ has a local maximum.

(3) If $f(0) = 0$, then for all reals x , $f(x) \geq -\frac{1}{3}$.

(4) If $f(0) = 0$, then the value of $\left| \int_{-1}^1 \frac{f'(x)}{4} dx \right| = \frac{2}{3}$.

Questions with Answer Keys

MathonGo

Q48

The correct statement(s) is/are -

- (1) The line of intersection of planes $\vec{r} \cdot \vec{n}_1 = q_1, \vec{r} \cdot \vec{n}_2 = q_2$ and $\vec{r} \cdot \vec{n}_3 = q_3, \vec{r} \cdot \vec{n}_4 = q_4$ are perpendicular if $(\vec{n}_1 \cdot \vec{n}_3)(\vec{n}_2 \cdot \vec{n}_4) = (\vec{n}_1 \cdot \vec{n}_4)(\vec{n}_2 \cdot \vec{n}_3)$.
- (2) If three distinct planes $\vec{r} \cdot \vec{n}_1 = q_1, \vec{r} \cdot \vec{n}_2 = q_2, \vec{r} \cdot \vec{n}_3 = q_3$ intersect in a line which is contained by the plane $\vec{r} \cdot \vec{n}_4 = q_4$, then $[\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_4] \vec{n}_3 = [\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_3] \vec{n}_4$.
- (3) If four distinct planes $\vec{r} \cdot \vec{n}_1 = q_1, \vec{r} \cdot \vec{n}_2 = q_2, \vec{r} \cdot \vec{n}_3 = q_3$ and $\vec{r} \cdot \vec{n}_4 = q_4$ intersect in a line, then $[\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_4] \vec{n}_3 = [\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_3] \vec{n}_4$.
- (4) If a plane contains line of intersection of planes $\vec{r} \cdot \vec{n}_1 = q_1, \vec{r} \cdot \vec{n}_2 = q_2$ and is parallel to line of intersection of planes $\vec{r} \cdot \vec{n}_3 = q_3, \vec{r} \cdot \vec{n}_4 = q_4$ then $[\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_4] \vec{n}_3 = [\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_3] \vec{n}_4$.

Q49

For polynomials of the form $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ with $a_i \in \{-1, 1\}, (i = 0, 1, 2, \dots, n)$ which has all roots real then.

- (1) the maximum value of n is less than 7
- (2) the maximum value of n is more than 5
- (3) Number of such polynomials is less than 10
- (4) Number of such polynomials is more than 5

Q50

Which of the followings has the value equal to the integral $\int_{1/e}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)}$?

- (1) 1
- (2) 2
- (3) $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} \cos t^2 dt}{x \sin x}$
- (4) $\lim_{n \rightarrow \infty} \left\{ \tan \frac{\pi}{2n} \tan \frac{2\pi}{2n} \tan \frac{3\pi}{2n} \dots \dots \tan \frac{n\pi}{2n} \right\}^{1/n}$

Q51

Questions with Answer Keys

MathonGo

Paragraph:

Consider, sum of the series $\sum_{0 \leq i < j \leq n} f(i)f(j)$

In the given summation, i and j are not independent.

In the sum of series $\sum_{i=1}^n \sum_{j=1}^n f(i) f(j) = \sum_{i=1}^n \left(f(i) \left(\sum_{j=1}^n f(j) \right) \right)$ i and j are independent. In this summation, three types of terms occur, those when $i < j$, $i > j$ and $i = j$.

Also, sum of terms when $i < j$ is equal to the sum of the terms when $i > j$ if $f(i)$ and $f(j)$ are symmetrical.

So, in that case

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n f(i)f(j) &= \sum_{0 \leq i < j \leq n} f(i)f(j) \\ &\quad + \sum_{0 \leq i < j \leq n} f(i)f(j) + \sum_{i=j} f(i)f(j) \\ &= 2 \sum_{0 \leq i < j \leq n} f(i)f(j) + \sum_{i=j} f(i)f(j) \\ \Rightarrow \sum_{0 \leq i < j \leq n} f(i)f(j) &= \frac{\sum_{i=0}^n \sum_{j=0}^n f(i)f(j) - \sum_{i=j} f(i)f(j)}{2} \end{aligned}$$

When $f(i)$ and $f(j)$ are not symmetrical, we find the sum by listing all the terms.

When $f(i)$ and $f(j)$ are not symmetrical, we find the sum by listing all the terms.

Question:

$\sum_{0 \leq i < j \leq n} \sum_{i=0}^n C_i^n C_j^n$ is equal to

- (1) $\frac{2^{2n} - 2^n C_n}{2}$
- (2) $\frac{2^{2n} + 2^n C_n}{2}$
- (3) $\frac{2^{2n} - n C_n}{2}$
- (4) $\frac{2^{2n} + n C_n}{2}$

Q52**Paragraph:**

Consider, sum of the series $\sum_{0 \leq i < j \leq n} f(i)f(j)$

In the given summation, i and j are not independent.

In the sum of series $\sum_{i=1}^n \sum_{j=1}^n f(i) f(j) = \sum_{i=1}^n \left(f(i) \left(\sum_{j=1}^n f(j) \right) \right)$ i and j are independent. In this

Questions with Answer Keys

MathonGo

summation, three types of terms occur, those when $i < j$, $i > j$ and $i = j$.

Also, sum of terms when $i < j$ is equal to the sum of the terms when $i > j$ if $f(i)$ and $f(j)$ are symmetrical.

So, in that case

$$\begin{aligned}\sum_{i=1}^n \sum_{j=1}^n f(i)f(j) &= \sum_{0 \leq i < j \leq n} f(i)f(j) \\ &\quad + \sum_{0 \leq i < j \leq n} f(i)f(j) + \sum_{i=j} f(i)f(j) \\ &= 2 \sum_{0 \leq i < j \leq n} f(i)f(j) + \sum_{i=j} f(i)f(j) \\ \Rightarrow \sum_{0 \leq i < j \leq n} f(i)f(j) &= \frac{\sum_{i=0}^n \sum_{j=0}^n f(i)f(j) - \sum_{i=j} f(i)f(j)}{2}\end{aligned}$$

When $f(i)$ and $f(j)$ are not symmetrical, we find the sum by listing all the terms.

When $f(i)$ and $f(j)$ are not symmetrical, we find the sum by listing all the terms.

Question:

$$\sum_{0 \leq i \leq j \leq n} \sum ({}^nC_i + {}^nC_j)$$

$$(1) (n+2)2^n$$

$$(2) (n)2^n$$

$$(3) (n-1)2^n$$

$$(4) (n+1)2^n - 1$$

Q53**Paragraph:**

If $f(x) = \frac{x}{x^2+x+1}$, then

On the basis of above information, answer the following :

Question:

If $\int f(\tan x) dx = x - \frac{2}{\sqrt{3}} \tan^{-1}(g(x)) + c$, then for $x \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$ range of $g(x)$ is -

$$(1) \left(\frac{1}{\sqrt{3}} - 2, \frac{1}{\sqrt{3}} + 2\right)$$

$$(2) (-\sqrt{3}, \sqrt{3})$$

$$(3) \left(-\frac{\sqrt{3}}{4-\sqrt{3}}, \frac{\sqrt{3}}{4+\sqrt{3}}\right)$$

(4) none of these

Q54

Paragraph:

If $f(x) = \frac{x}{x^2+x+1}$, then

On the basis of above information, answer the following :

Question:

If $\int f(x^2) dx + \int f\left(\frac{1}{x^2}\right) d\left(\frac{1}{x}\right) = \lambda \ln \left| \frac{x^2-x+1}{x^2+x+1} \right| + c$, then λ is -

(1) 1

(2) -1

(3) $\frac{1}{2}$

(4) $-\frac{1}{2}$

Answer Key

Q1 (3)	Q2 (3)	Q3 (3)	Q4 (2)
Q5 (1)	Q6 (2)	Q7 (2) (3) (4)	Q8 (1) (3) (4)
Q9 (1) (3)	Q10 (2) (4)	Q11 (1)	Q12 (1) (3)
Q13 (1) (3)	Q14 (1) (3)	Q15 (4)	Q16 (3)
Q17 (3)	Q18 (2)	Q19 (2)	Q20 (1)
Q21 (2)	Q22 (1)	Q23 (4)	Q24 (2)
Q25 (1) (2) (3) (4)	Q26 (1) (2) (4)	Q27 (2) (3)	Q28 (1) (2)
Q29 (2) (4)	Q30 (1)	Q31 (2) (3)	Q32 (1) (4)
Q33 (1)	Q34 (3)	Q35 (1)	Q36 (4)
Q37 (4)	Q38 (3)	Q39 (3)	Q40 (4)
Q41 (4)	Q42 (4)	Q43 (1) (2) (3)	Q44 (1) (2)
Q45 (1) (2) (4)	Q46 (1) (3)	Q47 (1) (3) (4)	Q48 (1) (2) (3) (4)
Q49 (1) (3) (4)	Q50 (1) (3) (4)	Q51 (1)	Q52 (2)
Q53 (1)	Q54 (3)		